

Problem Set 4

Quantitative Economics, Fall 2025

January 15, 2026

This problem set consists of two problems. Submit your solutions by Feb. 13th, 11:59 PM. You can work in teams of up to five students.

Problem 1: Consumption–Savings with Borrowing Constraints

Consider an infinite-horizon consumption–savings problem with idiosyncratic income risk. The household chooses consumption c and next period assets a' . You will solve it in two ways. The purpose of the exercise is to introduce you to a simple trick that can improve the accuracy of methods based on linear interpolation.

Preferences. Period utility is CRRA:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \quad \gamma > 0,$$

with the convention $u(c) = \log c$ when $\gamma = 1$.

Income. Labor income z follows a log AR(1) process:

$$\log z' = \rho \log z + \varepsilon', \quad \varepsilon' \sim N(0, \sigma_\varepsilon^2).$$

You will discretize this process with the Rouwenhorst method.

Budget and constraints. Given current assets a and income z ,

$$\begin{aligned} a' &= Ra + z - c, \\ c &\geq 0, \\ a' &\geq \underline{a}. \end{aligned}$$

Bellman equation.

$$V(a, z) = \max_{c \in [0, Ra + z - \underline{a}]} \{u(c) + \beta \mathbb{E}_z V(a', z')\},$$

where $a' = Ra + z - c$.

Parameterization Use:

- Risk aversion: $\gamma \in \{2, 10\}$ (solve for both)
- Discount factor: $\beta = 0.99$
- Interest rate: $R = 1.010$ for $\gamma = 2$, $R = 1.008$ for $\gamma = 10$.

- Income persistence: $\rho = 0.90$
- Income volatility: $\sigma_\varepsilon = 0.20\sqrt{1 - \rho^2}$
- Income discretization: Rouwenhorst with $N_z = 11$ states
- Asset grid: $a \in [\underline{a}, 500]$ with $N = 100$ grid points. Set the borrowing limit $\underline{a} = -0.6 \frac{z_{\min}}{R-1}$. This corresponds to 0.6 of the natural borrowing limit.

Create a grid by first generating ω_i for $i = 1, \dots, N$ that are evenly spaced in $[0, 1]$ and then setting $a_i = \underline{a} + (a_{\max} - \underline{a})\omega_i^\theta$ with $\theta = 3$. Here $z_{\min}$ is the minimum income state in your discretized Markov chain.

Observe that the largest debt that the agent can service is $z_{\min}/(R-1)$.

- Convergence: iterate until $\max_{i,j} \frac{|V^{n+1}(a_i, z_j) - V^n(a_i, z_j)|}{1 + |V^n(a_i, z_j)|} < 10^{-7}$.

Steps to follow:

You will solve the model using **two numerical implementations** and compare outcomes for $\gamma = 2$ vs $\gamma = 10$ (and the corresponding R values).

In both cases you will use linear interpolation to evaluate the value function at off-grid points and the Brent's method to solve the one-dimensional maximization problem.

The first method is standard value function iteration (VFI) with linear interpolation. The second method rewrites the Bellman equation using a monotone transformation of the value function (the “recursive CES trick”) before applying VFI with linear interpolation:

Feel free to interpolate the expected value function instead of the value function directly.

$$V(a, z) = \max_c \left[(1 - \beta)c^{1-\gamma} + \beta \mathbb{E}_z \left((a', z')^{1-\gamma} \right) \right]^{1/(1-\gamma)}.$$

The point is that this represents same preferences, but makes the value function less curved. This is good, because it means that it is easier to approximate with linear interpolation.

For both methods, for each $\gamma \in \{2, 10\}$ report:

- number of iterations and runtime,
- plots of $c(a, z)$ and $a'(a, z)$ for three income states (lowest, median, highest). Create also zoomed-in plots to highlight behavior near the borrowing constraint.
- plots of Euler equation errors over the asset grid for the three income states. As above, create zoomed-in plots to highlight behavior near the borrowing constraint.

Simulate behavior for $T = 10,000$ periods using the discretized Markov chain for z and your policy functions. Initialize at $a_0 = 0$ and z_0 equal to the median income state. Discard the first 500 periods.

Remember to use the same random seed for both methods and both γ values so that the income shocks are identical across cases.

Report:

- Mean assets $\bar{a}$ and standard deviation σ_a .
- Mean consumption $\bar{c}$ and standard deviation σ_c .
- Largest and smallest values of assets and consumption in the simulation.
- fraction of time at the borrowing constraint (define “at” as $a' \leq \underline{a} + 10^{-3}$),
- Autocorrelations $\text{corr}(a_t, a_{t-1})$ and $\text{corr}(c_t, c_{t-1})$ and correlations of consumption and assets with income $\text{corr}(c_t, z_t)$ and $\text{corr}(a_t, z_t)$.
- Provide a table comparing all four cases.
- Mean, median, 10th and 90th percentiles of the Euler equation errors over the simulation for each case.
- For each γ , plot $\{a_t, c_t\}$ over the first 100 post burn-in periods (overlay the two methods). Plot also the corresponding income process $\{z_t\}$. Plot the Euler equation errors over the same periods.

Problem 2: One-time Transfer and MPCs

Use the model from Problem 1 with $\gamma = 2$. If you found significant differences between the two solution methods in Problem 1, use the method that produced more accurate results (based on Euler equation errors). If you found that the quality of results was not good enough with either method, improve your code until you get satisfactory accuracy before proceeding. This might involve increasing the number of grid points, changing the asset grid, or other adjustments.

Individual marginal propensities to consume (MPCs) Let $c(a, z)$ be the consumption policy function from your baseline solution. Define the (local) MPC for transfer size $\Delta > 0$ as

$$\text{MPC}(a, z; \Delta) \equiv \frac{c(a + \Delta, z) - c(a, z)}{\Delta}.$$

- Compute $\text{MPC}(a, z; \Delta)$ on the grid for $\Delta \in \{0.01, 0.1, 0.5, 1.0, 2.0\}$.
- Plot MPC vs. a for (i) the lowest, (ii) median, and (iii) highest income states.
- Briefly discuss how MPCs vary with wealth, income, and transfer size.

To evaluate $c(a + \Delta, z)$, you will need to interpolate your policy function since $a + \Delta$ will generally be off-grid.

Aggregate response You will now study the effects of a one-time transfer of size equal to 5% of aggregate consumption.

Compute the stationary distribution $\lambda^*(a, z)$ implied by your policy $a'(a, z)$ and the income transition matrix Q .

Let aggregate consumption in the steady state be

$$C^* = \sum_{a,z} c(a, z) \lambda^*(a, z).$$

Let the transfer size be $\Delta = 0.05C^*$. Define the post-transfer distribution at $t = 0$ by shifting assets: households start period 0 with assets $a + \Delta$ instead of a . Carefully implement the shift on the discrete grid (e.g., by interpolation / mass-splitting). Compute

$$C^0 = \sum_{a,z} c(a, z) \lambda^0(a, z).$$

The aggregate MPC is $\text{MPC}^0 = (C^0 - C^*) / \Delta$.

For $t \geq 0$, propagate the distribution forward using the same transition matrix and the savings policy a' . For $t = 0, 1, \dots, 50$, compute

$$\lambda^{t+1} = \mathcal{F}(\lambda^t), \quad C^t = \sum_{a,z} c(a, z) \lambda^t(a, z),$$

where $\mathcal{F}$ is the mapping implied by a' and Q .

1. Plot the impulse response $(C^t - C^*) / C^*$ for $t = 0, \dots, 50$.
2. Compute cumulative MPCs

$$\text{MPC}^H = \frac{\sum_{t=0}^{H-1} (C^t - C^*)}{\Delta}$$

for $H \in \{0, 1, 4, 8, 12, 20\}$.

3. Report the fraction of the transfer spent by $H = 12$.

Note that this is a one-time transfer, everyone expects that there will be no future transfers. In addition, we assume that R is kept unchanged.